

RAILWAY OPS AGENTIC COPILOT

A closed-loop, multi-agent system for disruption handling in high-density railway corridors — decisions are constraint-verified before an explanation is ever generated.

● ALL SIGNALS CLEAR

AMD RADEON · ROCm LOCAL INFERENCE

01 THE PROBLEM

- Disruption handling in dense rail corridors is complex, high-stakes, and today largely error-prone.
- Dispatchers fall back on manual judgement — or on black-box LLMs that can hallucinate infeasible schedules.
- What's missing is a decision layer that is explainable, verifiable, and fast enough for real-time use.

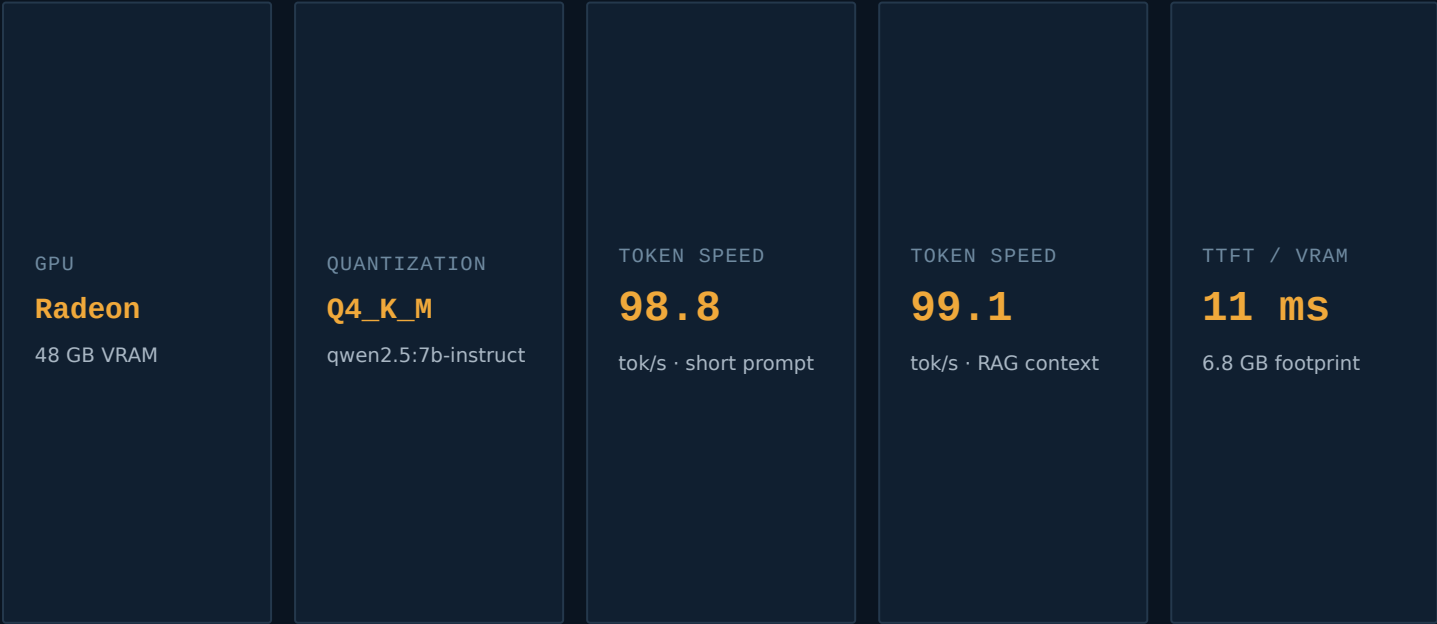
Our answer: *let solvers decide, let rules verify, and let a local LLM only narrate — never invent.*

02 CLOSED-LOOP MULTI-AGENT PIPELINE



The LLM sits **last** in the loop, downstream of a verified plan — it explains the decision, it does not make it.

03 ROCM PERFORMANCE — MEASURED LIVE



// local inference only — no cloud API in the decision or explanation path

04 REAL DATASET & VALIDATION



Seeded from real operational timing profiles rather than synthetic schedules, so the **Verifier Agent**'s headway and clearance checks are exercised against genuine corridor density.